

The Regress of Uncertainty and the Forecasting Paradox

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Abstract

We show that epistemic uncertainty—our ignorance about our own ignorance—inevitably thickens statistical tails. Any claim of precise risk carries a margin of error, and that margin itself is uncertain, in an infinite regress of doubt. This “errors-on-errors” mechanism rules out thin-tailed certainty: predictive laws must be heavier-tailed than their in-sample counterparts. The result is the Forecasting Paradox: the future is structurally more extreme than the past. This insight collapses branching scenarios into a single heavy-tailed forecast, with direct implications for risk management, scientific modeling, and AI safety.

1 The Regress Argument and the Structure of Ignorance

All quantitative models are maps of reality, not reality itself. This gap between representation and truth is the source of model risk and a fundamental subject of statistical inference [56]. While a long and distinguished tradition of philosophical inquiry into probability—including Laplace, Ramsey, Keynes, De Finetti, von Mises, and Jeffreys [8, 11, 12, 37, 67]—has grappled with the nature of belief and evidence, its deepest epistemological questions have often remained at the periphery of applied risk management. Yet epistemology is a central, not peripheral, element of the field [63, 60]. Any rigorous attempt at inference must eventually confront the foundational queries: *How do you know what you know? How certain are you of that knowledge?* This paper argues that the structure of our answer to these questions has direct, unavoidable, and mathematically potent consequences. We go further by considering higher orders of this doubt: *How certain are you of your certainty?*

In philosophy and decision theory this concern has long been expressed through the distinction between risk and uncertainty (and “potential surprises”), already emphasized by Knight [35] and Shackle [59], and later formalized by scholars like Halpern [26]. In the policy sciences, the same challenge is framed as “deep uncertainty”—a condition where analysts cannot agree on models, probability distributions, or even on the structure of causal relationships [24, 39, 44]. In climate economics, Weitzman [68], among others [38, 47], highlighted how ambiguity about sensitivity parameters leads to fat-tailed climate and welfare risks (leading to the so-called “dismal theorem”). In machine learning, the problem resurfaces in the tendency of modern AI systems to be overconfident: despite good predictive accuracy (on the short run), they often lack calibrated and reliable estimates of their own uncertainty [2, 21, 31, 51]. These strands of literature, spanning philosophy, economics, risk management, and computer science, converge on the same central insight: *our ignorance about our ignorance structurally reshapes predictive distributions.*

Our investigation begins with a simple, tautological observation about estimation. Any estimate, whether statistical or otherwise, is by definition an imprecise proxy for a true value. As such, it must possess an error rate; otherwise, it would be a certainty, if not a perfect reflection of reality. Here, however, we enter a chain of nested doubts: an epistemic regress. The estimated error rate is itself an estimate. The methods used to determine it are themselves imprecise. Therefore, the estimate of the error rate must also have an error rate. This recursion has no logical endpoint. There is no unimpeachable final layer, no bedrock of perfect knowledge upon which our models can rest. By a regress argument, our inability to account for this cascade of “errors on errors” fosters a deep model uncertainty whose full, recursive structure is often truncated at the first order and remains largely underappreciated in the standard literature [16, 66].

The consequences of this regress are not merely philosophical; they have a direct and powerful effect on probability itself. The mechanism is simple: uncertainty about a probability inflates the probability of tail events. To see this, assume a forecaster tells you the probability of a catastrophic event is *exactly* zero (thus violating Cromwell’s rule [5]). The crucial epistemological question is not whether the statement is true, but “How do you know?” If the answer is, “I estimated it,” then the claim of absolute certainty collapses. An estimate implies an error. The “true” probability must lie in a range above zero, however small (even if the error is small, it cannot be symmetric around zero, as probabilities are bounded below). Whenever our knowledge is not perfect, and thus with the exclusion of statements that are logically true or false, epistemic uncertainty structurally forbids claims of impossibility and forces us to assign higher probabilities to rare events than our base models would suggest.

A similar concern has been emphasized in applied risk management,

where the underestimation of tail probabilities has direct consequences. In finance, model risk is often discussed in terms of misspecified dynamics or wrong volatility inputs [15, 46]; in epidemiology and climate science, ensemble forecasting practices attempt to deal with disagreement across models. Yet these approaches typically treat the ensemble itself as given, avoiding the "ensemble of ensembles" problem. Our approach highlights that this hierarchy of uncertainty is the true, deep structure of model risk: uncertainty propagates recursively, and the structure of this regress thickens tails across domains. Practitioners often halt this regress after the first step, treating their own statements of uncertainty—such as confidence intervals—as if they were themselves known with certainty. This is a profound mistake that leads to the monumental underestimation of risk from higher moments [18, 61].

This challenge becomes acute in forecasting [53], where uncertainty multiplies. Standard methods for projecting the future, such as the counterfactual analysis of Lewis [41] or its business-school version, "scenario analysis," suffer from a combinatorial explosion. Each step forward in time branches into multiple possible outcomes, which in turn branch further, creating an intractable tree of futures that grows at a rate of at least 2^n for n steps. One of the principal contributions of this paper is to show that this seemingly intractable branching can be tamed. We present a method to structure the regress of errors analytically, collapsing the exploding tree of counterfactuals into a single, well-defined probability distribution. By parameterizing the rate of "error on error," we can vary our degree of epistemic doubt and perform sensitivity analyses on our own ignorance.

In what follows, we show how, by taking this epistemic argument to its mathematical conclusion, fatter tails necessarily emerge from layered uncertainty. We begin with a maximally "tame," thin-tailed world represented by the Gaussian distribution. By systematically perturbing its scale parameter, that is, by introducing explicit, nested layers of doubt about its "true" value, we show analytically how tail risk increases until the distribution becomes robustly heavy-tailed. Our argument is that this is not a mere mathematical curiosity, but a reflection of the true state of the world. As innumerable philosophers like de Finetti, Bradley, Nietzsche, and Russell have observed, our claims to objective truth are an invention [13, 6, 49, 57]. Our knowledge is and will always be an imperfect approximation—"adequatio intellectus et rei," as Isaac Israeli wrote [4, 40]. Real-world risk is therefore necessarily fatter-tailed than in our idealized models.

The mathematical techniques employed in this paper are well established. The novelty of this work lies not in isolated formulas, but in the unifying epistemological principle they collectively illustrate. By casting the infinite regress of uncertainty into a tractable analytical structure, we show that many disparate phenomena—financial risk, climate uncertainty, AI overconfidence, epidemiological forecasts—are all manifestations of the same mechanism:

each layer of doubt structurally thickens the tails of predictive laws. The binomial-tree representation is for instance a simple device to visualize this compounding mechanism, but the lasting contribution is the general epistemic framework that collapses branching scenarios into a universal law of fattened tails.

The paper is structured as follows. We begin in Section 2 by rigorously defining the terminology related to distribution tails. In Section 3, we introduce our core model of layered uncertainty, which we then analyze under the assumptions of a constant error rate and a decaying error rate. In Section 4, we use a Central Limit Theorem argument to provide a first generalization of the results, and provide a tractable analytical approximation in Section 5. In Section 6, we show that this is a universal mechanism, proving that scale uncertainty generates fat tails for any light-tailed baseline. In Section 7, we discuss the profound consequences of our findings, formalizing the Forecasting Paradox and exploring its applications in finance, AI safety, and scientific modeling. Section 8 concludes.

2 Basic terminology about tails

Let X be a random variable with cumulative distribution function F and survival function $\bar{F}$. Following [48], and to clarify some analyses we will perform, it is useful to distinguish rigorously among various notions of tail heaviness.

A *light-tailed* distribution has a finite moment generating function in a neighborhood of the origin, i.e.

$$M_X(t) = \mathbb{E}[e^{tX}] < \infty \quad \text{for some } t > 0.$$

Equivalently, its survival function satisfies

$$\limsup_{x \rightarrow \infty} \frac{\log \bar{F}(x)}{x} < 0,$$

so the tail decays at least exponentially fast. Gaussian, Exponential, and Gamma laws are light-tailed.

A *heavy-tailed* distribution is one with $M_X(t) = \infty$ for all $t > 0$. Equivalently,

$$\lim_{x \rightarrow \infty} e^{cx} \bar{F}(x) = \infty \quad \text{for every } c > 0,$$

so its tail decays strictly slower than any exponential e^{-cx} . Within heavy tails, finer classes are distinguished: long tails, subexponential tails and fat tails.

F is *long-tailed* if for every fixed $t \geq 0$,

$$\frac{\bar{F}(x+t)}{\bar{F}(x)} \rightarrow 1 \quad (x \rightarrow \infty).$$

For a nonnegative (absolutely continuous) X with unbounded support, long-tailedness implies that the hazard rate $q(x) \rightarrow 0$ and that the mean residual life $m(x) = \mathbb{E}[X - x \mid X > x]$ diverges as $x \rightarrow \infty$. Long-tailed distributions satisfy the so-called *explosion principle*: once a large value is observed, the tail beyond it remains asymptotically undiminished over any fixed further stretch [48].

F is said *subexponential* if

$$\bar{F}^{*2}(x) \sim 2\bar{F}(x) \quad (x \rightarrow \infty),$$

where $\bar{F}^{*2}$ denotes the survival function of the convolution $F * F$, i.e. the law of $X_1 + X_2$ for i.i.d. $X_1, X_2 \sim F$. The implication is simple: in the extreme, the maximum dominates the sum. This is the *catastrophe principle*: a single large shock outweighs many moderate ones. If you think of a financial portfolio, it is like saying that one single large loss accounts for most of the total loss of the portfolio over a given time horizon [18].

Finally, F is *fat-tailed* (or regularly varying) if

$$\mathbb{P}(X > x) = x^{-\alpha}L(x), \quad \alpha > 0,$$

with $L(x)$ being a slowly varying function at infinity. Formally, this means that for any constant $t > 0$, $\lim_{x \rightarrow \infty} L(tx)/L(x) = 1$. Intuitively, this implies that the tail of the distribution, asymptotically, behaves like a power law [33]. The most important property of fat-tailed distributions is that moments of order $k \geq \alpha$ do not exist (i.e., they diverge) [9], making their empirical counterparts unreliable for inference [65].

One can show [48] that Fat tails $\subset$ Subexponential tails $\subset$ Long tails $\subset$ Heavy tails. In words: every fat tail is also subexponential, long, and heavy, but not vice versa.

Important: In the following, borrowing from the parlance of practitioners, and unless otherwise stated or strictly necessary, we will often use “fatter-tailed” relative to a baseline to mean larger excess kurtosis and asymptotically larger upper-tail probabilities, i.e., riskier than the comparison case, without committing to whether the law is strictly fat-tailed, subexponential, or long-tailed.

3 Layering Uncertainties

Take a rather standard probability distribution, say the Normal, which falls in the thin-tailed class [18, 48]. Assume that its dispersion parameter, the standard deviation σ , is to be estimated following some statistical procedure to get $\hat{\sigma}$. Such an estimate will nevertheless have a certain error, a rate of epistemic uncertainty, which can be expressed with another measure

of dispersion: a dispersion on dispersion, paraphrasing the “volatility on volatility” of option operators [15, 17, 61]. This makes particular sense in the real world, where the asymptotic assumptions [58] usually made in mathematical statistics do not hold [61], and where every model and estimation approach is subsumed under a subjective choice [13].

Let $\phi(x; \mu, \sigma)$ be the probability density function (pdf) of a normally distributed random variable X with known mean μ and unknown standard deviation σ . To account for the error in estimating σ , we can introduce a density $f_1(\hat{\sigma}; \bar{\sigma}_1, \sigma_1)$ over $\mathbb{R}^+$, where σ_1 represents the scale parameter of $\hat{\sigma}$ under f_1 , and $\bar{\sigma}_1 = \sigma$ its expected value. We are thus assuming that $\hat{\sigma}$ is an unbiased estimator of σ , but our treatment could also be adapted to the weaker case of consistency [58]. In other words, the estimated volatility $\hat{\sigma}$ is the realization of a random quantity, representing the true value of σ with an error term.

The unconditional law of X is thus no longer that of a simple Normal distribution, but it corresponds to the integral of $\phi(x; \mu, \hat{\sigma})$ across all possible values of $\hat{\sigma}$ according to $f_1(\hat{\sigma}; \bar{\sigma}_1, \sigma_1)$. This is known as a scale mixture of normals [3, 69], and in symbols one has

$$g_1(x) = \int_0^\infty \phi(x; \mu, \hat{\sigma}) f_1(\hat{\sigma}; \bar{\sigma}_1, \sigma_1) d\hat{\sigma}. \quad (1)$$

Depending on the choice of f_1 , that in Bayesian terms would define an a priori, $g_1(x)$ can take different functional forms.

Now, what if σ_1 itself is subject to errors? As observed before, there is no obligation to stop at Equation (1): one can keep nesting uncertainties into higher orders, with the dispersion of the dispersion of the dispersion, and so forth. There is no reason to have certainty anywhere in the process.

For $i = 1, \dots, n$, set $\bar{\sigma}_i = \mathbb{E}[\hat{\sigma}_i]$, with $\bar{\sigma}_1 = \sigma$, and for each layer of uncertainty i define a density $f_i(\hat{\sigma}_i; \bar{\sigma}_i, \sigma_i)$, with $\hat{\sigma}_1 = \hat{\sigma}$. Generalizing to n uncertainty layers, one then gets that the unconditional law of X is now

$$g_n(x) = \int_0^\infty \int_0^\infty \cdots \int_0^\infty \phi(x; \mu, \hat{\sigma}_n) f_1(\hat{\sigma}_1; \bar{\sigma}_1, \sigma_1) \cdots f_n(\hat{\sigma}_n; \bar{\sigma}_n, \sigma_n) d\hat{\sigma}_1 d\hat{\sigma}_2 \cdots d\hat{\sigma}_n.$$

where $\hat{\sigma}_n$ denotes the “effective” volatility after n nested perturbations.

This approach is clearly parameter-heavy and computationally intensive, as it requires the specification of all the subordinated densities f_i for the different uncertainty layers and the resolution of a possibly very complicated integral.

Let us consider a simpler version of the problem, by playing with a basic multiplicative process à la Gibrat [23], in which the estimated σ is perturbed at each level of uncertainty i by dichotomic alternatives: overestimation or underestimation. We take the probability of overestimation to be p_i , while that of underestimation is $q_i = 1 - p_i$.

Let us start from the true parameter σ , and let us assume that its estimate is equal to

$$\hat{\sigma} = \begin{cases} \sigma(1 + \epsilon_1) & \text{with probability } p_1, \\ \sigma(1 - \epsilon_1) & \text{with probability } q_1, \end{cases}$$

where $\epsilon_1 \in [0, 1)$ is an error rate (for example it could represent the proportional mean absolute deviation [62]).

Equation (1) thus becomes

$$g_1(x) = p_1 \phi(x; \mu, \sigma(1 + \epsilon_1)) + q_1 \phi(x; \mu, \sigma(1 - \epsilon_1)).$$

Now, just to simplify notation, but without any loss of generality, hypothesize that, for $i = 1, \dots, n$, overestimation and underestimation are equally likely, i.e. $p_i = q_i = \frac{1}{2}$. Clearly one has that

$$g_1(x) = \frac{1}{2} (\phi(x; \mu, \sigma(1 + \epsilon_1)) + \phi(x; \mu, \sigma(1 - \epsilon_1))).$$

Assume now that the same type of uncertainty affects the error rate ϵ_1 , so that we can introduce $\epsilon_2 \in [0, 1)$ and define the element $(1 \pm \epsilon_1)(1 \pm \epsilon_2)$. Figure 1 gives a tree representation of the uncertainty over a few layers.

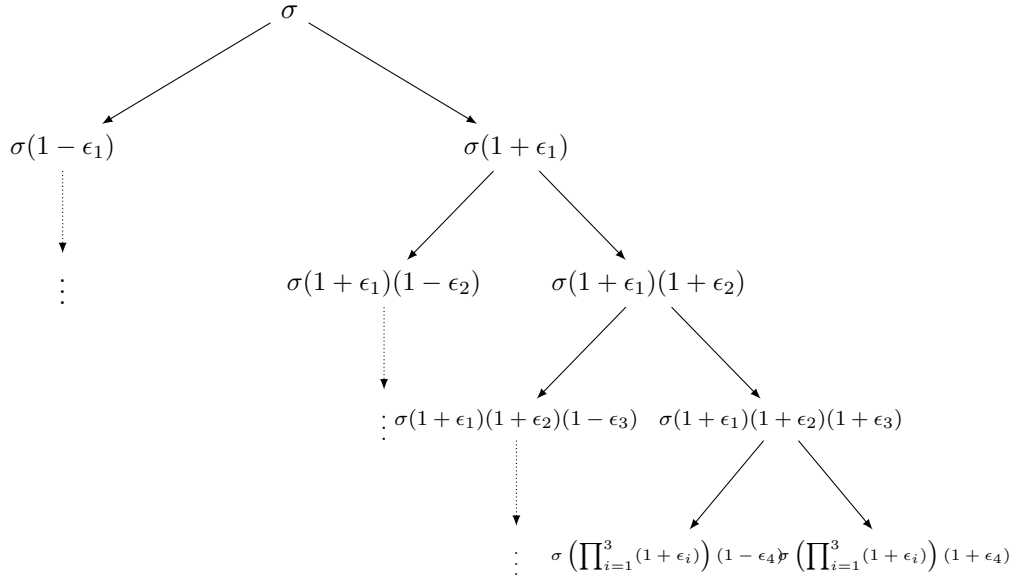


Figure 1: Tree representation of the layers of uncertainty. The diagram shows the progression to the fourth layer along a single path to illustrate the recursive mechanism, with vertical dots indicating the omitted branches.

With two layers of uncertainty the law of X thus becomes

$$g_2(x) = \frac{1}{4} \left[\phi(x; \mu, \sigma(1 + \epsilon_1)(1 + \epsilon_2)) + \phi(x; \mu, \sigma(1 - \epsilon_1)(1 + \epsilon_2)) \right. \\ \left. + \phi(x; \mu, \sigma(1 + \epsilon_1)(1 - \epsilon_2)) + \phi(x; \mu, \sigma(1 - \epsilon_1)(1 - \epsilon_2)) \right].$$

While at the n -th layer, we recursively get

$$g_n(x) = 2^{-n} \sum_{i=1}^{2^n} \phi \left(x; \mu, \sigma M_i^{(n)} \right), \quad (2)$$

where $M_i^{(n)}$ is the i -th entry of the vector

$$\mathbf{M}^{(n)} = \left[\prod_{j=1}^n (1 + \epsilon_j T_{i,j}) \right]_{i=1}^{2^n},$$

with $T_{i,j} \in \{-1, 1\}$ enumerating all sign patterns. For example, for $n = 3$, one convenient ordering is

$$\mathbf{T} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & -1 & 1 \\ -1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{M}^{(3)} = \begin{bmatrix} (1 - \epsilon_1)(1 - \epsilon_2)(1 - \epsilon_3) \\ (1 - \epsilon_1)(1 - \epsilon_2)(1 + \epsilon_3) \\ (1 - \epsilon_1)(1 + \epsilon_2)(1 - \epsilon_3) \\ (1 - \epsilon_1)(1 + \epsilon_2)(1 + \epsilon_3) \\ (1 + \epsilon_1)(1 - \epsilon_2)(1 - \epsilon_3) \\ (1 + \epsilon_1)(1 - \epsilon_2)(1 + \epsilon_3) \\ (1 + \epsilon_1)(1 + \epsilon_2)(1 - \epsilon_3) \\ (1 + \epsilon_1)(1 + \epsilon_2)(1 + \epsilon_3) \end{bmatrix}.$$

Once again, it is important to stress that the various error rates ϵ_i are not sampling errors, but rather projections of error rates into the future. They are, to repeat, of epistemic nature.

Equation (2) can be analyzed from different perspectives. In what follows we will discuss two relevant hypotheses regarding the error rates ϵ_i : constant error rates and decaying error rates. We omit increasing error rates, as they trivially only exacerbate the situation. Finally, we answer an important question for practitioners: do we really need $n \rightarrow \infty$? The answer is no, showing that we are not just philosophizing.

Remark 1 (Relation to Bayesian Inference). *Our approach shares mathematical tools with hierarchical Bayesian modeling, as both involve integrating over parameter uncertainty. However, our work is distinguished by its epistemological starting point. A standard Bayesian analysis models first-order uncertainty (a prior on a parameter) and may extend to second or third orders in a hierarchical model. The hierarchy, however, is typically truncated at a level deemed sufficient for the practical problem at hand.*

Our central thesis is that, from a logical standpoint, there is no justifiable stopping point. Any layer where the analyst assumes fixed hyperparameters represents a claim to perfect knowledge, violating the principle of the regress. Our contribution is not to propose a deeper hierarchical model, but to formalize the structure of this infinite regress itself, showing that its necessary consequence—regardless of the specific distributions chosen—is the thickening of predictive tails. We are thus providing a philosophical and mathematical foundation for why predictive distributions must be structurally fatter-tailed than any model with a finite, arbitrarily chosen level of uncertainty.

Remark 2 (Relation to the Renormalization Group). *The iterative layering of uncertainty on a scale parameter is, in a narrow mathematical sense, similar to a one-dimensional Renormalization Group (RG) flow [70]. In statistical physics, RG is a powerful tool for understanding how a system’s properties behave across different physical scales by averaging out small-scale details. The emergence of heavy-tailed distributions in our model can be seen as mathematically parallel to an RG flow approaching a fixed point under a multiplicative cascade [32, 36, 43].*

However, the analogy breaks down at the conceptual level, as our framework is more general in its philosophical scope and domain of application. The distinction is epistemological:

- *RG is a tool for describing objective system dynamics. It applies to physical systems that possess specific properties like scale invariance or are near a critical point. The flow it describes is an objective property of the system as the physical scale of observation changes.*
- *The Epistemic Regress is a principle of subjective knowledge. It requires no physical assumptions about the system being modeled and applies to the parameters of any model. The flow is not across physical scales but through the analyst’s own layers of ignorance. A system’s scale may be fixed; it is the scale of our uncertainty that evolves nevertheless.*

Our contribution, therefore, is not to re-derive the mathematics of scale invariance: it is to identify the Regress of Uncertainty as a universal principle of knowledge and forecasting. While RG describes a special ontic property that some systems have, our regress describes a universal property that all analysts (and all of us) have when facing an uncertain world. Our regress thus renormalizes credence rather than coupling, evolving through orders of doubt rather than spatial scales. Or, in other words, the RG flows through matter, while the epistemic regress flows through mind: its fixed points are limits of understanding, not of energy.

Remark 3 (Asymmetry). *In the previous derivations we assumed, for simplicity, that upward and downward shocks are equally likely, i.e. $p_i = q_i = \frac{1}{2}$. This symmetry is not necessary: one can allow for general probabilities p_i*

and $q_i = 1 - p_i$ at each layer. In such a case the distribution $g_n(x)$ becomes a weighted sum of Normal densities where the weights are no longer binomially symmetric but follow the corresponding multinomial law induced by the sequence $\{p_i\}$. As a result, the mixture acquires a systematic bias: when $p_i > q_i$, the dispersion parameter is more likely to be inflated, shifting mass towards fatter right tails; conversely, if $p_i < q_i$, underestimation dominates, attenuating tail risk relative to the symmetric case. Nevertheless, the distribution still exhibits fatter tails than the baseline Gaussian. Hence, asymmetry in the probabilities of over- vs. under-estimation not only magnifies or attenuates tail risk, but also introduces skewness into the unconditional law of X .

3.1 Hypothesis 1: Constant error rate

Assume that $\epsilon_1 = \epsilon_2 = \dots = \epsilon_n = \epsilon$, i.e. we have a constant error rate at each layer of uncertainty. What we can immediately observe is that matrix $\mathbf{M}$ collapses into a standard binomial tree for the dispersion at level n , so that

$$g_n(x) = 2^{-n} \sum_{j=0}^n \binom{n}{j} \phi\left(x; \mu, \sigma(1 + \epsilon)^j (1 - \epsilon)^{n-j}\right).$$

Because of the linearity of the sum, when ϵ is constant, we can use the binomial distribution to weight the moments of X , when taking n layers of epistemic uncertainty. One can easily check that the first four raw moments read as

$$\begin{aligned} \mu'_1 &= \mu, \\ \mu'_2 &= \mu^2 + \sigma^2 (1 + \epsilon^2)^n, \\ \mu'_3 &= \mu^3 + 3\mu\sigma^2 (1 + \epsilon^2)^n, \\ \mu'_4 &= \mu^4 + 6\mu^2\sigma^2 (1 + \epsilon^2)^n + 3\sigma^4 (1 + 6\epsilon^2 + \epsilon^4)^n. \end{aligned}$$

From these, one can then obtain the following notable moments [52]:

$$\begin{aligned} \text{Mean : } \mu, \quad \text{Variance : } \sigma^2 (1 + \epsilon^2)^n, \\ \text{Skewness : } 0, \quad \text{Excess kurtosis : } 3 \left(\frac{(1 + 6\epsilon^2 + \epsilon^4)^n}{(1 + \epsilon^2)^{2n}} - 1 \right). \end{aligned}$$

It is then interesting to measure the effect of n on the thickness of the tails of X . The obvious effect, as per Figure 3, is the rise of tail risk.

Fix n and consider the exceedance probability of X over a given threshold K , i.e. the tail of X , when ϵ is constant. One clearly has

$$\mathbb{P}(X \geq K) = \sum_{j=0}^n 2^{-n-1} \binom{n}{j} \operatorname{erfc}\left(\frac{K - \mu}{\sqrt{2} \sigma (1 + \epsilon)^j (1 - \epsilon)^{n-j}}\right),$$

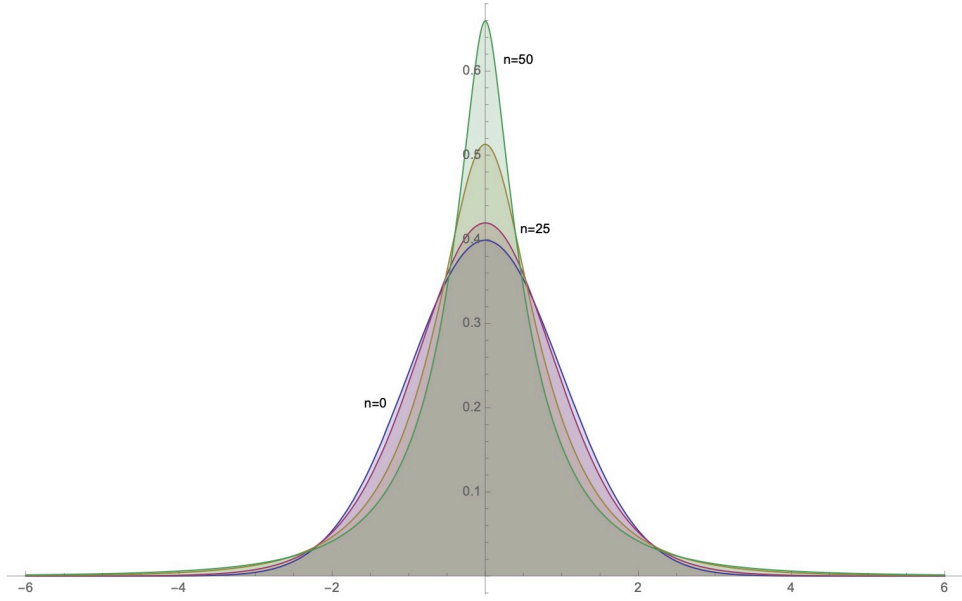


Figure 2: Examples of the density of X , when $\mu = 0$, $\sigma = 1$ and $\epsilon = 0.1$, for $n = 0, 5, 10$. The larger n , the fatter the tail.

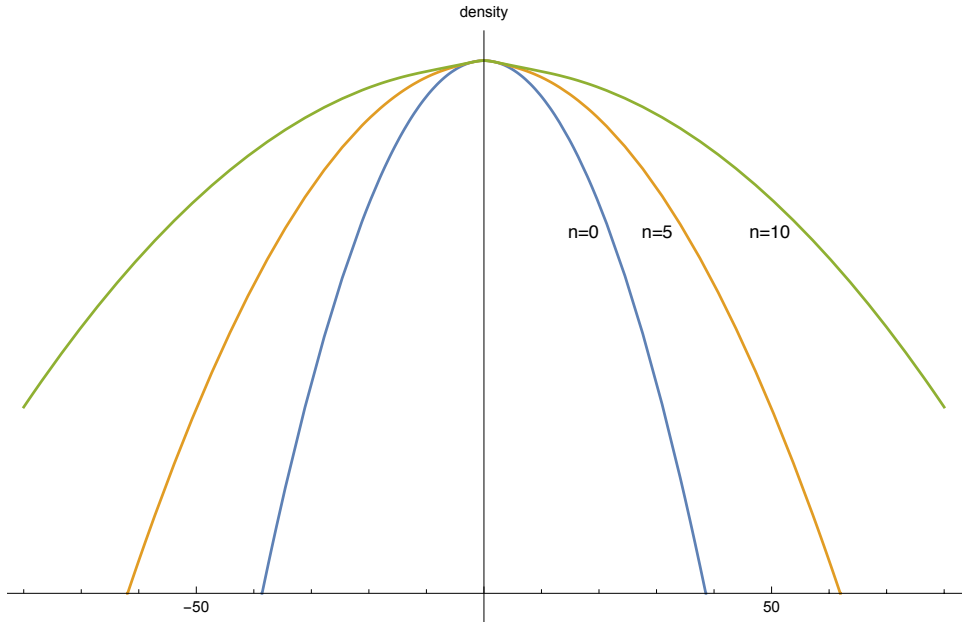


Figure 3: Log-plot of the density of X for $\epsilon = 0.1$ and different values of n . As n grows the tails decay much more slowly than Gaussian; over finite ranges the curves may appear approximately straight on a log-scale.

where $\operatorname{erfc}(z) = 1 - \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt$ is the complementary error function.

The results in Tables 1 and 2 quantify the dramatic impact of layered uncertainty. Even with a small epistemic error of 1% ($\epsilon = 0.01$ in Table 1), the probability of a 10-sigma event with 25 layers of uncertainty becomes over 3,300 times larger than what the baseline Normal model would suggest.

When the epistemic error is more substantial, say 10% ($\epsilon = 0.1$ in Table 2), the effect becomes explosive. The probability of the same 10-sigma event is magnified by a factor of 3.6×10^{18} —a number so vast it transforms an event previously dismissed as impossible into a conceivable threat. This demonstrates that ignoring the regress of uncertainty is not a minor oversight, but a source of massive, quantifiable underestimation of risk.

Table 1: Ratio of the exceedance probability of X over that of a Normal, for $\epsilon = 0.01$.

Layers (n)	Threshold K		
	$K = 3$	$K = 5$	$K = 10$
5	1.017	1.156	7.57
10	1.035	1.327	45.19
15	1.052	1.515	221.53
20	1.069	1.720	922.24
25	1.086	1.943	3347.0

Table 2: Ratio of the exceedance probability of X over that of a Normal, for $\epsilon = 0.1$.

Layers (n)	Threshold K		
	$K = 3$	$K = 5$	$K = 10$
5	2.747	146.10	1.09×10^{12}
10	4.437	805.99	8.998×10^{15}
15	5.985	1980.80	2.214×10^{17}
20	7.383	3529.41	1.210×10^{18}
25	8.642	5321.36	3.623×10^{18}

3.2 Hypothesis 2: Decaying error rates

As observed before, one may have (actually one needs to have) *a priori* reasons to stop the regress argument and take n to be finite. For example one could assume that the error rates vanish as the number of layers increases, so that $\epsilon_i \geq \epsilon_j$ for $i < j$, and ϵ_i tends to 0 when i approaches a given n . In this case, one can show that the higher moments tend to be capped, and the tail of X less extreme, yet riskier than what one could naively think.

Take a value $\kappa \in [0, 1]$ and fix ϵ_1 . Then, for $i = 2, \dots, n$, hypothesize that $\epsilon_i = \kappa \epsilon_{i-1}$, so that $\epsilon_n = \kappa^n \epsilon_1$. For what concerns X , without loss of generality, set $\mu = 0$. With $n = 2$, the variance of X becomes

$$\sigma^2 (1 + \epsilon_1^2) (1 + \kappa^2 \epsilon_1^2).$$

For $n = 3$ we get

$$\sigma^2 (1 + \epsilon_1^2) (1 + \kappa^2 \epsilon_1^2) (1 + \kappa^4 \epsilon_1^2).$$

For a generic n the variance is

$$\sigma^2 \prod_{i=0}^{n-1} (1 + \epsilon_1^2 \kappa^{2i}) = \sigma^2 [-\epsilon_1^2; \kappa^2]_n, \quad (3)$$

where $[a; q]_n = \prod_{i=0}^{n-1} (1 - aq^i)$ is the q -Pochhammer symbol.

Going on computing moments, for the fourth central moment of X , one gets for example

$$3\sigma^4 \prod_{i=0}^{n-1} (1 + 6\epsilon_1^2 \kappa^{2i} + \epsilon_1^4 \kappa^{4i}).$$

For $\kappa = 0.9$ and $\epsilon_1 = 0.2$, we get a variance of $1.23\sigma^2$, with a significant yet relatively benign convexity bias. And the limiting fourth central moment is $9.88\sigma^4$, more than 3 times that of a simple Normal, which is $3\sigma^4$. Such a number, even if finite—hence the corresponding scenario is not as extreme as before—definitely suggests a tail risk not to be ignored.

For values of κ in the vicinity of 1 and $\epsilon_1 \downarrow 0$, the fourth moment of X converges towards that of a Normal, closing the tails, as expected.

3.3 Do we really need infinity?

It is critical to note that the thickening of the tails does not require $n \rightarrow \infty$. The impact of epistemic uncertainty is immediate and significant even for very small values of n . A practitioner does not need to contemplate an infinite regress; acknowledging just one or two layers of doubt is enough to materially alter the risk profile. It is therefore totally understandable to put a cut-off somewhere for the layers of uncertainty, but such a decision should be taken *a priori* and motivated, in the philosophical sense. One should thus explicitly acknowledge model risk.

Let's consider a practical example. Assume a baseline model of a standard Normal distribution, $X \sim \mathcal{N}(0, 1)$, which represents the naive view with no parameter uncertainty ($n = 0$). Now, let a risk manager introduce a single layer of uncertainty ($n = 1$), acknowledging that their estimate of $\sigma = 1$ could be wrong by, say, $\epsilon = 20\%$.

- **For $n = 0$ (the naive model):** The probability of a 3-sigma event is $\mathbb{P}(X > 3) \approx 0.00135$, or about 1 in 740. The excess kurtosis is 0.
- **For $n = 1$ (one layer of doubt):** The distribution becomes a 50/50 mixture of $\mathcal{N}(0, 1.2^2)$ and $\mathcal{N}(0, 0.8^2)$. The probability of a 3-sigma event is now:

$$\begin{aligned}\mathbb{P}(X > 3) &= 0.5 \cdot \mathbb{P}(Z > 3/1.2) + 0.5 \cdot \mathbb{P}(Z > 3/0.8) \\ &= 0.5 \cdot \mathbb{P}(Z > 2.5) + 0.5 \cdot \mathbb{P}(Z > 3.75) \approx 0.00315,\end{aligned}$$

about 1 in 317.

Simply acknowledging one layer of 20% uncertainty about the volatility has more than doubled the probability of a 3-sigma event. The excess kurtosis has jumped from 0 to approximately 0.44. The effect is not asymptotic; it is a direct and quantifiable consequence of the very first step into the regress argument. Ignoring even a single layer of uncertainty thus constitutes a significant underestimation of risk.

4 A central limit theorem argument

We now discuss a central limit theorem argument for epistemic uncertainty as a generator of fatter tails and risk. For doing so, we introduce a more convenient representation of the normal distribution, which will also prove useful in Section 5.

Consider again the real-valued normal random variable X , with mean μ and standard deviation σ . Its density function is thus

$$\phi(x; \mu, \sigma) = \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sqrt{2\pi}\sigma}. \quad (4)$$

Without any loss of generality, let us set $\mu = 0$. Moreover let us re-parametrize Equation (4) in terms of a new parameter $\lambda = \frac{1}{\sigma^2}$, commonly called “precision” in Bayesian statistics [5]. The precision of a random variable X is nothing more than the reciprocal of its variance, and, as such, it is just another way of looking at variability (actually Gauss [22] originally defined the Normal distribution in terms of precision). From now on, we will therefore assume that X has density

$$\phi(x; \lambda) = \frac{\sqrt{\lambda} e^{-\frac{1}{2}(\lambda x^2)}}{\sqrt{2\pi}}. \quad (5)$$

Imagine now that we are provided with an estimate of λ , i.e. $\hat{\lambda}$, and take $\hat{\lambda}$ to be close enough to the true value of the precision parameter. Assuming that λ and $\hat{\lambda}$ are actually close is not necessary for our derivation, but we

want to be optimistic by considering a situation in which who estimates $\hat{\lambda}$ knows what she is doing, using an appropriate method, checking statistical significance, etc.

We can thus write

$$\lambda = \hat{\lambda}(1 + \epsilon_1), \quad (6)$$

where ϵ_1 is now a first-order random error term such that $\mathbb{E}[\epsilon_1] = 0$ and $\sigma^2(\epsilon_1) < \infty$. Apart from these assumptions on the first two moments, no other requirement is put on the probabilistic law of ϵ_1 .

Now, imagine that a second order error term ϵ_2 is defined on $1 + \epsilon_1$, and again assume that it has zero mean and finite variance. The term ϵ_2 may, as before, represent uncertainty about the way in which the quantity $1 + \epsilon_1$ was obtained. Equation (6) can thus be re-written as

$$\lambda = \hat{\lambda}(1 + \epsilon_1)(1 + \epsilon_2). \quad (7)$$

Iterating the error on error reasoning we can introduce a sequence $\{\epsilon_i\}_{i=1}^n$ such that $\mathbb{E}[\epsilon_i] = 0$ and $\sigma^2(\epsilon_i) \in [c, \infty)$, $c > 0$, so that we can write

$$\lambda = \hat{\lambda} \prod_{i=1}^n (1 + \epsilon_i). \quad (8)$$

For $n \rightarrow \infty$, Equation (8) represents our knowledge about the parameter λ , once we start from the estimate $\hat{\lambda}$ and we allow for epistemic uncertainty, in the form of multiplicative errors on errors. The lower value $c > 0$ for the variances of the error terms is meant to guarantee a minimum level of epistemic uncertainty at every level, and to simplify the application of the central limit argument below.

Now take the logs on both sides of Equation (8) to obtain

$$\log(\lambda) = \log(\hat{\lambda}) + \sum_{i=1}^n \log(1 + \epsilon_i). \quad (9)$$

If we assume that, for every $i = 1, \dots, n$, $|\epsilon_i|$ is small with respect to 1, we can introduce the approximation $\log(1 + \epsilon_i) \approx \epsilon_i$, and Equation (9) becomes

$$\log(\lambda) \approx \log(\hat{\lambda}) + \epsilon_1 + \dots + \epsilon_n. \quad (10)$$

To simplify treatment, let us assume that the error terms $\{\epsilon_i\}_{i=1}^n$ are independent from each other.¹ For n large, a straightforward application of the Central Limit Theorem (CLT) of Laplace–Liapounoff [19] tells us that $\log(\lambda)$ is approximately distributed as a Normal($\log \hat{\lambda}, S^2$), where $S^2 = \sum_{i=1}^n \sigma^2(\epsilon_i)$. This clearly implies that $\lambda \sim \text{Lognormal}(\log \hat{\lambda}, S^2)$, for $n \rightarrow \infty$. Notice that, for n large enough, we could also assume $\hat{\lambda}$ to be a random variable (with

¹In case of dependence, we can refer to one of the generalizations of the CLT [18, 19].

finite mean and variance), but still the limiting distribution of λ would be a Lognormal.

Epistemic doubt has thus a very relevant consequence from a statistical point of view. Using Bayesian terminology, the different layers of uncertainty represented by the sequence of random errors $\{\epsilon_i\}_{i=1}^n$ correspond to eliciting a Lognormal prior distribution on the precision parameter λ of the initial Normal distribution. This means that, in case of epistemic uncertainty, the actual marginal distribution of the random variable X is no longer a simple Normal, but a Compound Normal–Lognormal distribution, which we can represent as

$$g(x) = \int_0^\infty \underbrace{\frac{\sqrt{\lambda}}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\lambda x^2\right)}_{\text{Normal}} \cdot \underbrace{\frac{1}{\lambda S \sqrt{2\pi}} \exp\left(-\frac{(\log \lambda - \log \hat{\lambda})^2}{2S^2}\right)}_{\text{Lognormal}} d\lambda. \quad (11)$$

Despite its apparent simplicity, the integral in Equation (11) cannot be solved analytically, and no moment generating function can be defined (the normal-lognormal compound is indeed heavy-tailed). However, its moments can be obtained explicitly, providing a clear picture of how epistemic uncertainty inflates risk.

Let us derive the moments of X . Since the distribution is symmetric around $\mu = 0$, all odd moments are zero. For the even moments $\mu_{2m} = \mathbb{E}[X^{2m}]$, we can use the law of total expectation:

$$\mu_{2m} = \mathbb{E}[\mathbb{E}[X^{2m}|\lambda]], \quad \mathbb{E}[X^{2m}|\lambda] = \frac{(2m)!}{m! 2^m} \lambda^{-m}.$$

For $\lambda \sim \text{Lognormal}(\log \hat{\lambda}, S^2)$ and $k \in \mathbb{R}$, $\mathbb{E}[\lambda^k] = \hat{\lambda}^k \exp(k^2 S^2/2)$. Hence

$$\mu_{2m} = \frac{(2m)!}{m! 2^m} \hat{\lambda}^{-m} \exp\left(\frac{m^2 S^2}{2}\right).$$

By rescaling x we may normalize $\hat{\lambda} = 1$, yielding the simple forms

$$\text{Var}(X) = e^{S^2/2}, \quad \mu_4 = 3e^{2S^2}, \quad \mu_6 = 15e^{9S^2/2},$$

and an excess kurtosis of $3(e^{S^2} - 1)$.

5 An analytical approximation

The impossibility of solving Equation (11) can be bypassed by introducing an approximation to the Lognormal distribution on λ . The idea is to use a Gamma distribution to mimic the behavior of the lognormal prior on precision, also looking at tail behavior.

Both the Lognormal and the Gamma distribution are skewed distributions on $\mathbb{R}_+$ and share a convenient property: their coefficient of variation (CV) is constant with respect to location. For $\text{Lognormal}(\mu_L, S^2)$ the CV equals $\sqrt{e^{S^2} - 1}$; for $Y \sim \text{Gamma}(\alpha, \beta)$ with density

$$\frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\beta y}, \quad \alpha > 0, \beta > 0,$$

the CV is $1/\sqrt{\alpha}$.

From the point of view of extreme value theory [18], the Lognormal is subexponential (hence heavy-tailed) and in the Gumbel maximum domain of attraction [14, 18]; the Gamma is *light-tailed* (hence not subexponential) and also in the Gumbel domain. In the bulk they can be hard to distinguish [30, 45], but asymptotically the Lognormal tail dominates: for $\text{Gamma}(\alpha, \beta)$ (shape-rate) the hazard $h(x) \rightarrow \beta$ as $x \rightarrow \infty$, whereas for the Lognormal $h(x) \rightarrow 0$.

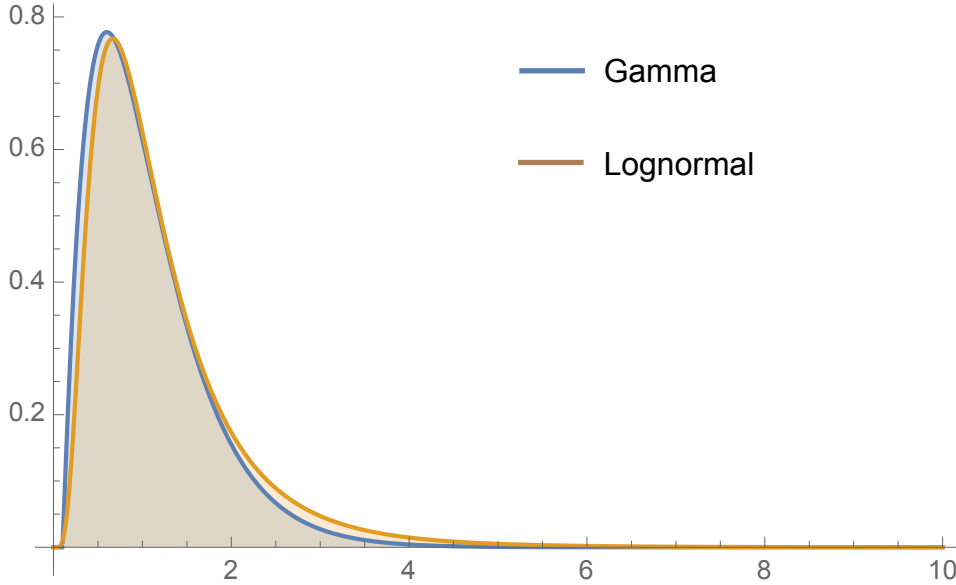


Figure 4: Comparison of the density functions of a Gamma and a Lognormal sharing the same coefficient of variation of 0.7.

In dealing with the precision parameter λ , moving from the Lognormal to the Gamma has a great advantage. A Normal distribution with known mean (for us $\mu = 0$) and Gamma-distributed precision parameter has an explicit closed form.

Let us rewrite Equation (11) by substituting the lognormal density with

an approximating $\text{Gamma}(\alpha, \beta)$ prior on λ :

$$\begin{aligned} g(x; \alpha, \beta) &= \int_0^\infty \frac{\sqrt{\lambda}}{\sqrt{2\pi}} e^{-\frac{1}{2}\lambda x^2} \cdot \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda} d\lambda \\ &= \frac{\beta^\alpha}{\Gamma(\alpha)\sqrt{2\pi}} \int_0^\infty \lambda^{\alpha+\frac{1}{2}-1} \exp\left(-\lambda\left(\beta + \frac{1}{2}x^2\right)\right) d\lambda. \end{aligned} \quad (12)$$

The integral above can be solved explicitly, yielding

$$g(x; \alpha, \beta) = \frac{\Gamma\left(\alpha + \frac{1}{2}\right)}{\Gamma(\alpha) \sqrt{2\pi\beta}} \left(1 + \frac{x^2}{2\beta}\right)^{-(\alpha+\frac{1}{2})}. \quad (13)$$

This is the density of a Student's t with

$$\nu = 2\alpha \quad \text{degrees of freedom, and scale} \quad s^2 = \beta/\alpha,$$

i.e.

$$g(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right) \sqrt{\pi\nu} s} \left(1 + \frac{x^2}{\nu s^2}\right)^{-\frac{\nu+1}{2}}.$$

Matching the coefficient of variation of $\text{Lognormal}(\log \hat{\lambda}, S^2)$ fixes

$$\alpha = \frac{1}{e^{S^2} - 1}, \quad \beta = \frac{e^{-S^2/2}}{e^{S^2} - 1} \hat{\lambda}^{-1}.$$

Interestingly, the Student's t distribution in Equation (13) is fat-tailed on both sides [14, 18], especially for small values of α . It is worth noting an apparent paradox in this result: we are mixing a Normal distribution (light-tailed) with a Gamma prior on its precision (which is also light-tailed), yet the result is robustly fat-tailed. This phenomenon highlights the power of the scale-mixing mechanism. The key is that the Gamma distribution is placed on the *precision* $\lambda = 1/\sigma^2$. A light-tailed prior on precision that allows for values arbitrarily close to zero implies a heavy-tailed prior on the *variance* σ^2 . It is precisely the possibility of drawing a near-zero precision (and thus a near-infinite variance) that generates the fat tails in the final predictive distribution.

Since α decreases in S^2 (the sum of the variances of the epistemic errors), the more doubts we have about the precision parameter λ , the more the resulting Student's t is fat-tailed, thus increasing tail risk. This is in line with the findings of Section 3.1.

Therefore, starting from a simple Normal distribution, by considering layers of epistemic uncertainty, we have obtained a thick-tailed predictive distribution with the same mean ($\mu = 0$), but capable of generating more extreme scenarios, and its tail behavior is a direct consequence of imprecision and ignorance.

Remark 4 (Conservative upper bound via Normal–Gamma). *Although the lognormal prior on λ has fatter tails than its Gamma approximation at the prior level, the resulting compound laws invert this ordering. The Normal–Gamma mixture is a Student’s t with power-law tails, asymptotically fatter than the Normal–Lognormal mixture (whose tail decays like $\exp\{-c(\log x)^2\}$). Thus the Normal–Gamma route provides a tractable and conservative upper bound for extreme tails.*

6 A Universal Mechanism: From Scale Uncertainty to Fat Tails

Using a simple example based on a Gaussian distribution, the preceding sections showed that layers of epistemic doubt about the standard deviation, modeled as a multiplicative regress, naturally lead to a heavy-tailed, compound distribution. But is this phenomenon specific to the Normal distribution, or does it reflect a more fundamental statistical law?

In this section, we demonstrate the latter. We generalize the mechanism and show that epistemic uncertainty about a model’s scale parameter is a universal generator of fatter tails. This principle provides a direct mathematical bridge from the philosophical concept of layered ignorance to the practical necessity of using heavy-tailed distributions for any realistic modeling, forecasting, or risk management endeavor. The mathematical foundation for this principle is the theory of scale mixtures, which states that mixing any light-tailed distribution with a heavy-tailed scale factor produces a new, heavy-tailed distribution, as formalized in the following proposition.

Proposition 1. *Let $Z \geq 0$ be independent of $S \geq 0$ and define $X := SZ$.*

- (i) *If S is subexponential on $\mathbb{R}_+$ and Z is light-tailed with $\mathbb{E}[Z^\delta] < \infty$ for some $\delta > 0$ (equivalently $M_Z(t) < \infty$ for some $t > 0$), and $\mathbb{P}(Z > 0) > 0$, then X is subexponential on $\mathbb{R}_+$. Moreover, for all $x > 0$,*

$$\bar{F}_X(x) = \mathbb{E}\left[\bar{F}_S\left(\frac{x}{Z}\right)\right].$$

- (ii) *If S is regularly varying with index $\alpha > 0$, and $\mathbb{E}[Z^{\alpha+\delta}] < \infty$ for some $\delta > 0$, then X is regularly varying with the same index α and*

$$\mathbb{P}(X > x) \sim \mathbb{E}[Z^\alpha] \mathbb{P}(S > x), \quad x \rightarrow \infty.$$

Proof. For (i): the identity $\bar{F}_X(x) = \mathbb{E}[\bar{F}_S(x/Z)]$ follows from independence and conditioning on Z . Closure of the subexponential class under multiplication by an independent light-tailed factor with a finite positive moment is standard; see, e.g., [20, Thm. 3.27 and Cor. 3.28].

For (ii): this is Breiman’s lemma [7] under the stated moment condition on Z . $\square$

Remark 5. *If the baseline is symmetric $Z_0 \stackrel{d}{=} -Z_0$, apply the theorem to $Z := |Z_0|$ to obtain the right-tail result for $|X| = S|Z_0|$, and then transfer it to two-sided tails by symmetry.*

Proposition 1 thus guarantees that our mechanism—representing ignorance via a random scale factor—produces heavy tails for any light-tailed phenomenon, not just the Normal. Our CLT argument leading to a Log-normal (subexponential) distribution for the scale factor is nothing but a specific instance of this universal principle.

Notice that Proposition 1 is not an abstract curiosity; it reveals the hidden structure of many well-known statistical models. Whenever a simple model is made more realistic by treating its rate, volatility, or scale parameter as unknown and variable, a fatter-tailed distribution emerges. The following examples illustrate this principle across different domains:

- **Exponential $\rightarrow$ Pareto:** An Exponential distribution with a fixed rate λ is light-tailed. If we express uncertainty about λ by assuming it follows a Gamma distribution, the resulting mixture distribution for the variable is a Lomax (Pareto Type II) distribution, which is fat-tailed.
- **Poisson $\rightarrow$ Negative Binomial:** A Poisson distribution is used for counts with a fixed rate μ . If we acknowledge that this rate is uncertain and model it with a Gamma distribution, the resulting compound distribution is the Negative Binomial. This distribution is famous for its "overdispersion" and fatter tail compared to the Poisson itself, making it far more suitable for modeling real-world count data.
- **Normal $\rightarrow$ Student's t:** As shown in Section 5, a Normal distribution (light-tailed) mixed with an Inverse-Gamma prior on its variance (equivalent to a Gamma prior on precision) results in a Student's t-distribution, the canonical example of a fat-tailed distribution in statistics.

These examples reveal a profound pattern: many canonical fat-tailed distributions can be interpreted as simple, light-tailed models whose scale has been randomized to account for uncertainty. The regress of uncertainty finds its mathematical expression in the operation of scale mixing. This provides a direct and robust path from epistemic doubt to the necessity of heavy-tailed models.

7 The Forecasting Paradox and its Consequences

The principle that layered epistemic uncertainty thickens tails is not a mere theoretical curiosity: it has profound and actionable consequences across

any domain that relies on statistical modeling and forecasting. Ignoring the regress of errors on errors leads to a systemic and dangerous underestimation of risk, brittleness in automated systems, and a false sense of certainty in scientific projections. Our framework thus provides a formal basis for a crucial principle of forecasting.

Principle 1 (The Forecasting Paradox). *The predictive distribution for out-of-sample data must be treated as having fatter tails than the descriptive model of in-sample data.*

The reason is now clear. A model fitted to past data (in-sample) is conditioned on a *point estimate* of its parameters, such as $\hat{\sigma}$. However, a forecast for the future (out-of-sample) cannot treat this estimate as perfect. One must account for the uncertainty surrounding $\hat{\sigma}$ itself. This is achieved by integrating over all possible values of the parameter, weighted by their probabilities—the very scale mixture operation that our paper proves is a generator of fat tails. Therefore, a responsible forecast must reflect this added layer of epistemic uncertainty, making it structurally more conservative than a historical description.

7.1 Applications in Finance and Quantitative Risk Management

The consequences for risk management are immediate. Standard models like Value-at-Risk (VaR) or Expected Shortfall (ES) often rely on fitting a distribution to historical data and then extrapolating its quantiles [34, 46]. In our opinion, this is insufficient and risky.

Let us discuss a practical example. An analyst wants to compute the one-day 99% Value-at-Risk (VaR) for a portfolio, based on a historical sample of daily returns. The standard procedure is as follows:

1. **The Standard (and Flawed) Approach:** The analyst calculates the historical standard deviation, obtaining a point estimate $\hat{\sigma} = 1.5\%$ per day. Assuming a mean-zero Gaussian distribution for simplicity [28], the 99% VaR is calculated as:

$$\text{VaR}_{\text{descriptive}} = 2.33 \times \hat{\sigma} = 2.33 \times 1.5\% = 3.495\%$$

This value is often directly used as a forecast of tomorrow’s risk [46]. This is a profound epistemological error in our view.

2. **The Robust Approach (The Regress of Uncertainty):** A prudent analyst, following our principle, must acknowledge that $\hat{\sigma} = 1.5\%$ is merely an estimate. To account for this, we model the volatility for the next day not as a point, but as a random variable. For simplicity, assume a minimal epistemic error of $\pm 10\%$, leading to a 50/50 mixture

of two scenarios: a "high volatility" world where $\sigma_H = 1.65\%$ and a "low volatility" world where $\sigma_L = 1.35\%$. The predictive distribution is now a mixture of two Gaussians.

3. **The Quantifiable Consequence:** The 99% VaR of this new, fatter-tailed distribution is not the average of the individual VaRs. It is the value x for which the probability of the loss being less than x is 99% under the mixture distribution. Formally, we must solve for $\text{VaR}_{\text{predictive}}$ in the equation:

$$0.5 \cdot \Phi\left(\frac{\text{VaR}_{\text{predictive}}}{\sigma_L}\right) + 0.5 \cdot \Phi\left(\frac{\text{VaR}_{\text{predictive}}}{\sigma_H}\right) = 0.99$$

where Φ is the standard Normal cumulative distribution function. Solving this numerically yields:

$$\text{VaR}_{\text{predictive}} \approx 3.56\%$$

This value is higher than the naive estimate of 3.495% due to convexity. The increase of 6.5 basis points, on a \$1 billion portfolio, translates into \$650,000 of additional underestimated risk capital for a single day. This gap represents the quantifiable cost of ignoring just the *first layer* of epistemic uncertainty.

This example provides a concrete demonstration of the Forecasting Paradox: the risk of the future is structurally greater than the risk observed in the past because the future must also contain our uncertainty about the parameters of the past. Standard methods systematically under-price this fundamental uncertainty.

To consider another situation, we all know that the Black–Scholes model’s assumption of constant volatility is famously flawed [18, 46, 61]. Models incorporating stochastic volatility (the “volatility of volatility”) are a step in the right direction [17, 46]. Our framework provides an epistemological justification for this, suggesting that the hierarchy does not stop. Uncertainty about the parameters of the volatility model itself must be considered, especially for long-dated options where such uncertainty compounds over time.

Finally, in market risk as well as in credit risk modeling, instead of relying on a handful of ad-hoc historical or imaginary scenarios [28, 46], our method allows for the systematic generation of a universe of scenarios. The parameters of model (n and ϵ_i , or S^2) can be seen as “dials of doubt.” By adjusting them, an institution can explore the consequences of varying degrees of model uncertainty, moving from simple sensitivity analysis to a full-blown distributional understanding of future risks.

7.2 Machine Learning and the Challenge of AI Safety

Modern machine learning, particularly deep learning, has achieved remarkable performance on many tasks, but this success is often undermined by a dangerous overconfidence [25, 42]. These models frequently produce point estimates with little to no reliable measure of their own uncertainty, a failure that becomes especially acute when they are faced with data different from what they were trained on [1].

Our framework is a form of uncertainty quantification. While methods like Bayesian Neural Networks (BNNs) place priors over model weights, our work suggests a deeper requirement: hierarchical models that account for uncertainty *in the hyperparameters of those priors*. A safe AI should not only provide a prediction but also a meta-assessment of its own certainty, derived from its nested model of self-doubt.

For high-stakes applications like autonomous driving or medical diagnostics, an AI that is “99.9% safe” is a liability if that certainty is brittle. The regress argument implies that true robustness comes from acknowledging the potential for errors in the model’s own construction. This leads to predictive distributions that are more cautious, assigning non-trivial probability to rare but catastrophic failures, which is the first step toward mitigating them.

Consider a practical example from an autonomous vehicle’s perception system. An AI model detects an object on the road and reports its classification with a very high point-estimate confidence: “99.5% probability this is a harmless plastic bag.” A standard, overconfident system would take this at face value, potentially leading to the decision to drive over it.

Our framework insists that this 99.5% figure is merely an estimate subject to its own regress of uncertainty. A robust AI must ask: What is my uncertainty about this confidence score? If the lighting is unusual or the object’s shape is ambiguous (i.e., the data is partially out-of-distribution), the “true” confidence is not a point but a distribution with fatter tails towards lower probability. The system might calculate that there is a non-trivial probability (say, 1%) that the object’s “true” classification confidence is actually below 50%. Faced with this ambiguity, the robust decision is not to trust the point estimate but to default to a safer action: slow down and avoid the object. This shows how formalizing the regress of uncertainty translates directly into more cautious and reliable AI behavior in safety-critical applications, in line with a rational precautionary principle [62, 64].

7.3 Humility in Scientific Modeling: Climate and Epidemiology

The same principle applies to complex scientific modeling where parameters are deeply uncertain [65]. During a risk event like a pandemic [10], different models produce a wide range of forecasts for variables like infection rates or

hospitalizations. This variance is a direct reflection of parameter uncertainty. A consolidated forecast should not be a simple average, but a formal mixture of the models’ outputs, which, as we have shown, results in a fatter-tailed distribution that better communicates the true range of possibilities to policymakers.

In climate science, long-term climate projections depend on parameters like “climate sensitivity,” for which we have only uncertain estimates [29, 47, 68]. The regress argument applies perfectly: there is an error on the estimate, an error on the error estimate, and so on. A true risk assessment of climate change must therefore be based on the fat-tailed distribution that arises from this layered uncertainty, forcing us to confront the non-negligible probability of extreme warming scenarios, even more extreme than commonly discussed [27, 29, 39, 50].

8 Conclusion

Ultimately, our approach provides a mathematical backbone to the philosophical wisdom of epistemic humility. As Rescher noted, taking estimates as objective truth is the greatest mistake a decision-maker can make [54]. This work explains the central point of *The Black Swan* [61]: one must treat the future as capable of delivering more frequent and impactful surprises than are found in the records of the past. The hindsight effect that characterizes “black swans” is the painful, delayed realization of having ignored the errors on errors.

By providing a mechanism to formalize our doubt, we can move from using models that project a fragile sense of certainty to those that build resilience. It is important to stress that the specific distributions used in this paper, such as the Normal, are merely pedagogical; the principle of the regress is universal, applying to any modeling endeavor where an estimate stands in for an unknown truth. Understanding that our own ignorance has a rigorous mathematical structure—one that leads directly to fatter tails—is the essential first step towards making decisions that are robust in the face of a deeply uncertain world.

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